

Compute the summations and products in 19–28.

5.1.38. $\sum_{k=1}^m \left(\frac{k}{k+1} \right) + \left(\frac{m+1}{m+2} \right)$

$$\sum_{k=1}^m \left(\frac{k}{k+1} \right) + \left(\frac{m+1}{m+2} \right) =$$

$$\left(\left(\frac{1}{1+1} \right) + \left(\frac{2}{2+1} \right) + \dots + \left(\frac{m}{m+1} \right) \right) + \left(\frac{m+1}{m+2} \right) =$$

$$\sum_{k=1}^{m+1} \left(\frac{k}{k+1} \right)$$

Solution: $\sum_{k=1}^m \left(\frac{k}{k+1} \right) + \left(\frac{m+1}{m+2} \right) = \sum_{k=1}^{m+1} \left(\frac{k}{k+1} \right)$